

Optimization problems in finance

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Optimization is essentially a practical tool and one principally used by non-mathematicians; in contrast, most research papers in optimization are written in a style that is only intelligible to a mathematician.
(Murray, 1972, p. vii)

In this chapter, we discuss financial optimization models in general: how models are set up, how they are solved, and how obtained solutions are evaluated. The chapter concludes with several examples of financial optimization models, some of which will be revisited in later chapters. Following William Murray’s quotation, all descriptions will be informal.

10.1 What to optimize?

An optimization model consists of an objective function (also called optimization criterion or goal function) and constraints. For all the applications discussed in later chapters, the objective function is scalar valued; it takes as arguments a vector x of decision variables, and the data. So our basic problem is

$$\underset{x}{\text{minimize}} \; f(x, \text{data}),$$

subject to constraints. “Minimize” is not restrictive: if we wanted to maximize, we would minimize $-f$. The model specification will be determined by considering and balancing different aspects:

- financial** The straightforward part. We define goals and the means to achieve them; and we make these notions precise. For an asset allocation model, for instance, we may need to decide how to define risk—for example, variability of returns, average losses, probability of failing to meet an obligation—or how to measure portfolio turnover.
- empirical** In finance, we are rarely interested in the past. (Even though we should be. Reading the works of Charles Kindleberger or John Kenneth Galbraith shows why.) All models deal with future—and hence unknown—quantities. So we need to forecast, estimate, simulate, or approximate these quantities. Building the model (the finance part) must not be separated from the empirical part; we may only deal with quantities that we can forecast to a sufficient degree. This is a strong, yet unavoidable, constraint on formulating the model.
- computational** There is a second constraint: the model must be solvable. Computational aspects of such models are the topic of this part of the book. We will argue that they are much less of a constraint than is sometimes thought.

We can think of model building as a meta-optimization in which we try to obtain the best possible results (or more realistically, good results; results that improve the current status) for our financial goals subject to restrictions that the model remains empirically meaningful and can be solved.

We will briefly discuss the ingredients of a model here. The objective function is often given by the problem at hand. In asset allocation, for instance, we may want a high return and low risk; when we calibrate a model, we want to choose parameter values so that the output of the model is “close” to observed quantities. Of course, these descriptions need to be made more precise. There are many ways to define what “close” means. We have to select, say, a specific norm of absolute or relative differences. In fact, many problems can be spelled out in different ways. When we estimate interest rate models, we may look at interest rates, but also at bond prices. When we look at options, we may work with prices, but also with implied volatilities. Bond prices are functions of interest rates; there is a bijective relationship between option prices and Black–Scholes-implied volatilities. Conceptually, different model formulations may be equivalent; numerically and, especially, empirically they are often not. Specific choices can make a difference, and they often do.

We can phrase this more to the point: can we directly optimize the quantities that we are interested in? The answer is No; at least, Not Always. A well-known example comes from portfolio selection. If we wanted to maximize return, we should not write it without safeguards into our objective function. The reason is that (i) we cannot predict future returns very well, and (ii) there is a cost involved in failing to correctly predict: the chosen portfolio performs poorly. Theory may help, but determining a good objective function—one that serves our financial goals—ultimately is an empirical task.

Most realistic problems have constraints. Constrained optimization is generally more difficult than the unconstrained case. Constraints may, like the objective function, be given by the problem at hand. In asset allocation we may have legal or regulatory restrictions on how to invest. Empirically, restrictions often have another purpose. They act as safeguards against optimization results that follow more from our specific data, rather than from the data-generating process. This can concern out-of-sample performance: in portfolio selection, there is much evidence that imposing maximum position sizes helps to improve performance. But constraints can also help to make estimated parameters interpretable. For example, variances cannot be negative, and probabilities must lie in the range $[0, 1]$. Yet when such quantities are the outputs of a numerical procedure, we are not guaranteed that these restrictions are observed, and so we need to make sure that our algorithms yield meaningful results.

10.2 Solving the model

10.2.1 Problems

The focus of this book is neither on financial nor empirical aspects of optimization models, but on their numerical solution. In the coming chapters we describe a number of optimization models that cannot be solved with standard methods, that is, models that pose difficult optimization problems. This is not meant presumptuously; it simply reflects the fact that many problems in finance are difficult to solve. Heuristic methods which we will describe below can help to obtain good solutions, as will be demonstrated in those coming chapters. A clarification is needed here: in optimization theory, the solution of a model is the optimum; it is not necessary to speak of “optimal solutions.” But that is not the case in practical applications. A solution here is rather the result obtained from a computer program. The quality of this solution will depend on the interplay between the problem and the chosen method (and chance).

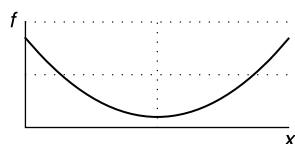
From a practical perspective, difficulty in solving a problem can be measured by the amount of computational resources required to compute a (good) solution. In computer science, there are the fields of complexity theory and analysis of algorithms that deal with the difficulty of problems, but results are often of limited use in applications. The practical efficiency of algorithms depends on their implementation; sometimes minor details can make huge differences. Often the only way to obtain results is to run computational experiments, which is exactly what we will do in later chapters.

But what makes an optimization problem difficult? For combinatorial problems, it is the problem size. Such problems have an exact solution method—just write down all possible solutions and pick the best one—but this approach is almost never feasible for realistic problem sizes. For continuous problems, difficulties arise when:

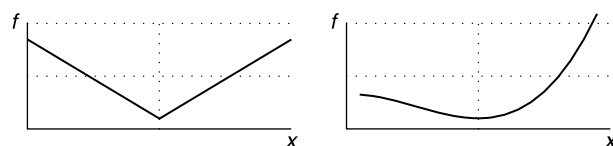
- The objective function is not smooth (e.g., has discontinuities) or is noisy. In either case, relying on the gradient to determine search directions may fail. An example is a function that needs to be evaluated for given arguments by a stochastic simulation or by another numerical procedure (e.g., a quadrature or a finite-difference scheme).
- The objective function has many local optima.

Practically, even in the continuous case we could apply complete enumeration. We could discretize the domain of the objective function and run a so-called grid search. But it is easy to see that this approach, just like for combinatorial problems, is not feasible in practice once the dimensionality of the model grows.

Let us look at some stylized examples of objective functions to make these problems clearer. First, we picture the nice case.

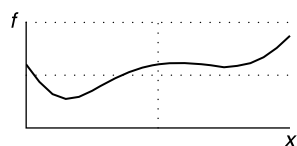


We only have one optimum, the function is smooth, and it is convex. But now imagine we have functions like this.



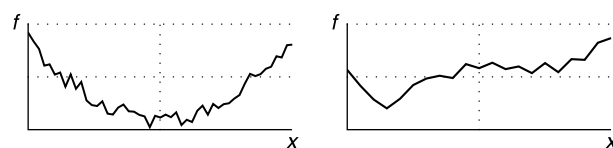
The left figure shows a kink in the objective function, so we cannot use an analytical derivative. The figure on the right has an inflection point, so the objective function is not convex. Newton's method for instance relies on f being globally convex as it will approximate f by a quadratic function, and then solve for the minimum of this function (see Fig. 11.9 on page 249). Already such apparently innocuous properties may cause trouble for standard optimization methods.

The functions pictured so far had one minimum, but what if the function looks like this:



We have two local minima, and one is clearly better than the other. Yet traditional methods will easily get trapped in such local (but suboptimal) minima.

As a final example, we may not be able to evaluate our objective function precisely, but only subject to noise.



These functions were stylized, but these problems really occur. Several real examples are presented below.

10.2.2 Classical methods and heuristics

To solve optimization models, many researchers and practitioners rely on what we call here standard or classical techniques. (You may have noticed that we have referred to “standard” methods several times before.) Classical methods are, for the purpose of this book, defined as methods that require convexity or at least well behaved objective functions since these techniques are often based on exploiting the derivatives of the objective function. Classical methods are mathematically well founded; numerically, there are powerful solvers available which can efficiently solve even large-scale instances of given problems. Methods that belong to this approach are, for instance, linear and quadratic programming. The efficiency and elegance of these methods comes at a cost, though, since considerable constraints are put on the problem formulation, that is, the functional form of the optimization criterion and the constraints. We often have to shape the problem such that it can be solved by these methods. Thus, the answer that the final model provides is a precise one, but often only to an approximative question.

An alternative approach that we will describe in this book is the use of heuristic optimization techniques. Heuristics are a relatively new development in optimization theory. Even though early examples date back to the 1950s or so, these methods have become practically relevant only in recent decades with the enormous growth in computing power. Heuristics aim at providing good and fast approximations to optimal solutions; the underlying theme of heuristics may thus be described as seeking approximative answers to exact questions. Heuristics have been shown to work well for problems that are completely infeasible for classical approaches (Michalewicz and Fogel, 2004). Conceptually, they are often simple; implementing them rarely requires high levels of mathematical sophistication or programming skills. Heuristics are flexible: we can easily add, remove, or change constraints, or modify the objective function. These advantages come at a cost as well, as the obtained solution is only a stochastic approximation, a random variable. However, such a stochastic solution may still be better than a poor deterministic one (which, even worse, we may not even recognize as such) or no solution at all when classical methods cannot be applied. In fact, for many practical purposes, the goal of optimization is far more modest than to find the truly best solution. Rather, any good solution, where good means an improvement of the status quo, is appreciated. And practically, we often get *very* good solutions. We give examples in later chapters.

Heuristics are not better optimization techniques than classical methods; the question is rather when to use what kind of method. If classical techniques can be applied, heuristic methods will practically always be less efficient. When, however, given problems do not fulfill the requirements of classical methods (and the number of such problems seems large), we suggest not to tailor the problem to the available optimization technique, but to choose an alternative—heuristic—technique for optimization.

We will discuss classical methods in Chapter 11. In Chapter 12, we then give a selective overview of heuristic techniques; the remaining chapters of Part III detail the implementation of these methods.

10.3 Evaluating solutions

Modeling is approximation. As described in Chapter 1, whenever we approximate we commit errors. Hence, a solution to our model will inherit these errors. In Chapter 1, we suggested to divide errors into two broad categories: model errors (i.e., empirical errors), and numerical errors. This book is mainly about the second type. In finance, however, the first type is much more important. And evaluating it is much more difficult. A practical approach is to compare different models of which some are “more accurate” than others. Accurate means that a model cannot only be solved with sufficient numerical precision, but that it is also economically meaningful. When a model is more accurate than another model, it has a lower empirical error, at least regarding some aspect of the original problem. Suppose we have two models that serve the same purpose. One model can be solved precisely, but is less accurate than a second model that can be only solved approximatively. We still can empirically test whether an only moderately good solution to the more accurate model provides a better answer to our real problem than the precise solution to the less accurate model. Again, an example from portfolio selection can illustrate this point. Markowitz (1959, Chapter 9)

compares two risk measures, variance and semi-variance, in terms of cost, convenience, familiarity, and desirability; he concludes that variance is superior in terms of cost, convenience, and familiarity. For variance we can compute exact solutions to the portfolio selection problem; for semi-variance we can only approximate the solution. Today, we can empirically test whether, even with an inexact solution for semi-variance, the gains in desirability outweigh the increased effort.

Even if we accept a model as true, the quality of the model's solution will be limited by the quality of the model's inputs, that is, data or parameter estimates. Appreciating these limits helps to decide how precise a solution we actually need. This decision is relevant for many problems in computational finance since we generally face a trade-off between the precision of a solution and the effort required (most visibly, computing time). Surely, the numerical precision with which we solve a model is important; we need reliable methods. Yet, empirically, there must be a sufficient precision for any given problem. Any improvement beyond this level cannot translate into gains regarding the actual problem any more; only in costs (increased computing time or development costs). Given the rather low empirical accuracy of many financial models, this required precision cannot be high. More specifically, when it comes to optimization we can decide whether we actually need an exact solution as promised by the application of classical methods, or whether a good solution provided by a heuristic is enough.

In principle, of course, there would seem to be little cost in computing precise solutions. Yet there is. First, highly precise or "exact" solutions will give us a sense of certainty that can never be justified by a model. Second, getting more-precise solutions will require more resources. This does not just mean more computing time, but also more time to develop a particular method. Grinold and Kahn (2008, pp. 284–285) give an example; they describe the implementation of algorithms for an asset selection problem—finding the portfolio with cardinality 50 that best tracks the S&P 500. This is a combinatorial problem for which an exact solution could never be computed (there are about 10^{70} possible portfolios), hence we need an approximation. A specialized algorithm took six months to be developed, but then delivered an approximative solution within seconds. As an alternative, a heuristic technique, a Genetic Algorithm, was tested. Implementation took two days; the algorithm found similar results, but also needed two days of computing time. A remark is in order: the example is from the 1990s. Today, the computing time of a Genetic Algorithm for such a problem would be of the order of minutes, perhaps seconds, on a standard PC. Researchers may have become cleverer since the 1990s, i.e. they may be developing models faster today; but it is unlikely that their improvement in performance matches that of computer technology.

A final consideration of the quality of a solution is the distinction between in-sample and out-of-sample. For financial models, this distinction is far more relevant than any numerical issue. More precise solutions will by definition appear better in-sample, but we need to test if this superiority is preserved out-of-sample. Given the difficulty we have in predicting or estimating financial quantities, it seems unlikely that highly precise solutions are necessary in financial models once we compare out-of-sample results. Again, we can set up an empirical test here. Suppose we have a model for which we can compute solutions with varying degrees of precision. Each solution can be evaluated by its in-sample objective function value (its in-sample fit), but each solution also corresponds to a certain out-of-sample quality. We can now sort our solutions by in-sample precision, and then see if out-of-sample quality is a roughly monotonous function of this precision, and if the slope of this function is economically meaningful.

To sum up this discussion: the numerical precision with which we solve a model matters, and in-sample tests can show whether our optimization routines work properly. The more relevant question though is to what extent this precision translates into economically meaningful solutions to our actual problems.

In the remaining part of this chapter, several examples for optimization problems that cannot be solved with classical methods will be given. This is not a survey, but a selection of problems to illustrate and motivate the use of heuristic methods in finance. Some of these problems will be revisited in later chapters. For more detailed studies, see for example Maringer (2005b). Many references to specific applications can be found in Schlottmann and Seese (2004) or Gilli et al. (2008).

10.4 Examples

Portfolio optimization with alternative risk measures

In the framework of modern portfolio optimization (Markowitz, 1952, 1959), a portfolio of assets is often characterized by a desired property, the reward, and something undesirable, the risk. Markowitz identified these two properties with the expectation and the variance of returns, respectively, hence the expression mean–variance optimization. By now, there exists a large body of evidence that financial asset returns are not normally distributed (see, for instance, Cont, 2001), thus, describing a portfolio by only its first two moments is often regarded as insufficient. Alternatives to mean–variance optimization have been proposed, in particular, replacing variance as the risk measure.

Assume an investor has wealth v_0 and wishes to invest for a fixed period of time. A given portfolio, as it comprises risky assets, maps into a distribution of wealth v_T at the end of the period. The optimization problem can be stated as follows

$$\begin{aligned} \min_x \quad & f(v_T(x)) \\ x_j^{\inf} \leq x_j \leq x_j^{\sup} \quad & j \in J \\ K_{\inf} \leq \# \{J\} \leq K_{\sup} \\ & \dots \text{and other constraints.} \end{aligned}$$

The objective function $f(\cdot)$ could be a risk measure or a combination of multiple objectives to be minimized. Candidates include the portfolio’s drawdown, partial moments, or whatever else we wish to optimize. The vector x stores the weights or the units (integer numbers) of assets held. $x_j^{\inf}$ and $x_j^{\sup}$ are vectors of minimum and maximum holding sizes, respectively, for those assets included in the portfolio (i.e., those in the set J). If short sales are allowed, this constraint could be modified to $x_j^{\inf} \leq |x_j| \leq x_j^{\sup}$. $K_{\inf}$ and $K_{\sup}$ are cardinality constraints which set a minimum and maximum number of assets to be in J . There may be restrictions on transaction costs (in any functional form) or turnover, lot size constraints (i.e., restrictions on the multiples of assets that can be traded), and exposure limits. We may also add constraints that, under certain market conditions, the portfolio needs to behave in a certain way (usually give a required minimum return).

Similar to this framework are index-tracking problems. Here investors try to replicate a predefined benchmark; see, for example, Gilli and K llezi (2002). This benchmark need not be a passive equity index. In the last few years, for instance, there have been attempts to replicate the returns of hedge funds; see Lo (2008).

Applying alternative risk measures generally necessitates using the empirical distribution of returns. (There is little advantage in minimizing kurtosis when stock returns are modeled by a Brownian motion.) The resulting optimization problem cannot be solved with classical methods (except for special cases like mean–variance optimization). To give an example, Fig. 10.1 shows the search space, that is, the values of the objective function that particular solutions map into, for a problem where f is the portfolio’s Value-at-Risk. The resulting surface is not convex and not smooth. Any search that requires a globally convex model, like a gradient-based method, will stop at the first local minimum encountered, if it arrives there at all.

For some objective functions, the optimization problem can be reformulated to be solved with classical methods; examples are Gaivoronski and Pflug (2005) or Rockafellar and Uryasev (2000); Chekhlov et al. (2005). But such solutions are problem specific and do not accommodate changes in the model formulation. How to use heuristics for portfolio selection will be discussed more thoroughly in Chapter 14.

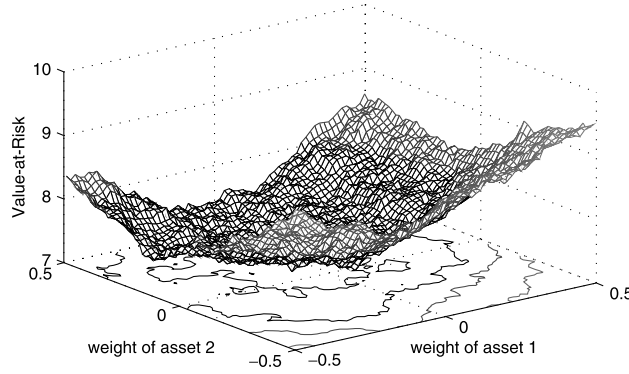


FIGURE 10.1 Objective function for Value-at-Risk.

Model selection

Linear regression is a widely used technique in finance. A common application are factor models where the returns of single assets are described as functions of other variables. Then

$$r = [F_1 \cdots F_k] \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} + \epsilon \quad (10.1)$$

with r being a vector of returns for a given asset, F_i are the vectors of factor realizations, β are the factor loadings, and ϵ captures the remaining variation. Such models are widely applied for instance to construct variance–covariance matrices or in attempts to forecast future returns. The factors F may be macroeconomic quantities or firm specific characteristics; alternatively, the analyst may use statistical factors, for instance extracted by principal component analysis. In practice, observable factors are often preferred since they are easier to interpret and can be better explained to clients. Given the vast amounts of financial data available, these factors may have to be picked from hundreds or thousands of available variables, since we may also consider lagged variables (Maringer, 2004). Model selection becomes a critical issue, as we often wish to use only a small number of k regressors from K possible ones, with $K \gg k$. We could use an information criterion, which penalizes additional regressors, as the objective function. Alternatively, techniques like cross-validation can be applied or the problem can be formulated as an in-sample fit maximization under the restriction that k is not greater than a (small) fixed number.

Robust/resistant regression

Empirical evidence has shown that the Capital Asset Pricing Model (CAPM) explains asset returns in the cross-section rather poorly (Fama and French, 1993, 2004). When we re-interpret the CAPM as a one-factor model (Luenberger, 1998, Chapter 8), however, the β estimates become useful measures of a stock’s general correlation with the market, which may be used to construct variance–covariance matrices (Chan et al., 1999).

The standard method to obtain parameter estimates in a linear regression is Least Squares. Least Squares has appealing theoretical and practical (numerical) properties, but obtained estimates are often unstable in the presence of extreme observations which are common in financial time series (Chan and Lakonishok, 1992; Knez and Ready, 1997; Genton and Ronchetti, 2008). Some earlier contributions in the finance literature suggested some form of shrinkage of extreme β estimates towards more reasonable levels, with different theoretical justifications (see for example Blume, 1971, or Vasicek, 1973). Alternatively, the use of robust or resistant estimation methods to obtain the regression parameters has been proposed (Chan and Lakonishok, 1992, Martin and Simin, 2003). Among possible regression criteria, high breakdown point estimators are often regarded as

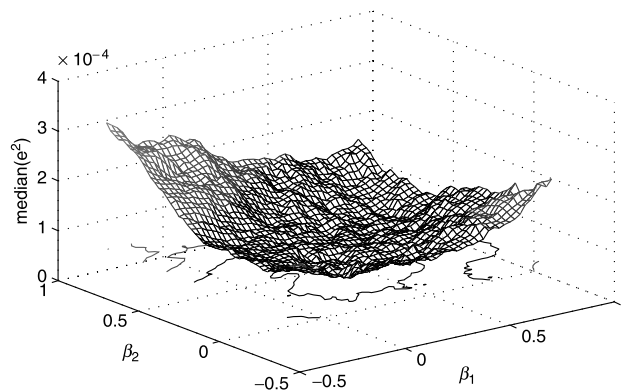


FIGURE 10.2 LMS objective function.

desirable. The breakdown point of an estimator is the smallest percentage of outliers that may cause the estimator to be affected by a bias. The Least Median of Squares (LMS) estimator, suggested by Rousseeuw (1984), ranks highly in this regard, as its breakdown point is 50%. (Note that Least Squares may equivalently be called Least Mean of Squares.)

Unfortunately, LMS regression leads to nonconvex optimization models, a particular search space for the simple model $y = \beta_1 + \beta_2 x + \epsilon$ is shown in Fig. 10.2. Estimation will be discussed in Chapter 16.

Agent-based models

Agent-based models (ABM) abandon the attempt to model markets and financial decisions with one representative agent (Kirman, 1992). This results in models that quickly become analytically intractable, hence researchers rely on computer simulations to obtain results. ABM are capable of producing many of the “stylized facts” actually observed in financial markets like volatility clustering, jumps or fat tails. For overviews on ABM in finance, see, for example, LeBaron (2000, 2006); see also Section 8.8.

Unfortunately, the conclusion of many studies stops at asserting that these models can in principle produce realistic market behavior when parameters (like preferences of agents) are specified appropriately. This leads to the question of what appropriate values should be like, and how different models compare with one another when it comes to explaining market facts.

Gilli and Winker (2003) suggest estimating the parameters of such models by indirect inference. This requires an auxiliary model that can easily be estimated, which in their case is simply a combination of several moments of the actual price data. A given set of parameters for the ABM is evaluated by measuring the distance between the average realized moments of the simulated series and the moments obtained from real data. This distance is then to be minimized by adjusting the parameters of the ABM. Winker et al. (2007) provide a more detailed analysis of objective functions for such problems.

Fig. 10.3 shows the resulting search space for a particular ABM (see Kirman, 1993). The objective function does not seem too irregular at all, but since the function was evaluated by a stochastic simulation of the model, it is noisy and does not allow for the application of classical methods.

Calibration of option-pricing models

Prices of options and other derivatives are modeled as functions of the underlying securities’ characteristics (Madan, 2001). Parameter values for such models are often obtained by solving inverse problems, that is, we try to obtain parameter values for which the model gives prices that are close to actual market prices. In case of the Black–Scholes model, only one parameter, volatility, needs to be specified, which can be done efficiently with Newton’s method (Manaster and Koehler, 1982). More recent option-pricing models (see, for instance, Bakshi et al., 1997, or Bates, 2003) aim to

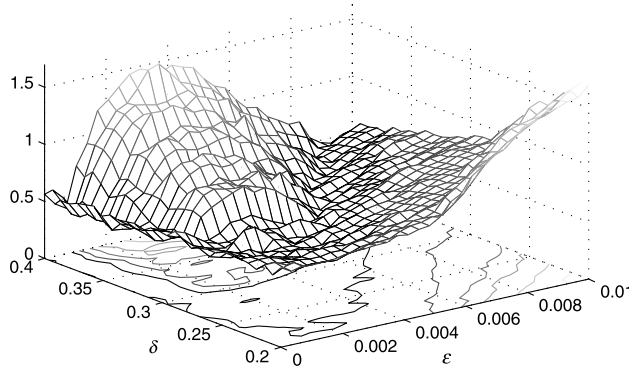


FIGURE 10.3 Simulated objective function for Kirman's model for two parameters.

generate prices that are consistent with the empirically observed implied volatility surface (Cont and da Fonseca, 2002). Calibrating these models requires setting more parameters, which leads to more difficult optimization problems.

One particular pricing model is that of Heston (1993); it is popular since it gives closed-form solutions. “Closed-form” is somewhat deceiving: pricing still requires numerical integration of a complex-valued function. Under the Heston model, the stock price (S) and variance (v) dynamics are described by

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{v_t} S_t dW_t^1 \\ dv_t &= \kappa(\theta - v_t)dt + \sigma\sqrt{v_t} dW_t^2 \end{aligned}$$

where the two Brownian motion processes are correlated, that is, $dW_t^1 dW_t^2 = \rho dt$. As can be seen from the second equation, volatility is mean-reverting to its long-run level θ with speed κ in the Heston model. In total, the model requires (under the risk-neutral measure) the specification of five parameters (Mikhailov and Nögel, 2003). Even though some of these parameters could be estimated from the time series of the underlying, the general approach to fit the model is to minimize the differences between the theoretical and observed prices. A possible objective function is, hence,

$$\min \sum_{n=1}^N w_n (C_n^H - C_n^M)^2$$

where N is the number of option quotes available, C^H and C^M are the theoretical and actual option prices, respectively, and w are weights (Hamida and Cont, 2005). Sometimes the optimization model also includes parameter restrictions, for example to enforce the parameters to be such that the volatility cannot become negative.

Fig. 10.4 shows the resulting objective function values for two parameters (volatility of volatility and mean-reversion speed) with the remaining parameters fixed. As can be seen, in certain parts of the parameter domain the resulting objective function is not too well behaved, hence standard methods may not find the global minimum. The Heston model is discussed in Chapter 17.

Calibration of yield structure models

The model of Nelson and Siegel (1987) and its extended version, introduced by Svensson (1994), are widely used to approximate the term structure of interest rates. Many central banks use the models to represent the spot and forward rates as functions of time to maturity; in several studies (e.g., Diebold and Li, 2006) the models have also been used for forecasting interest rates.

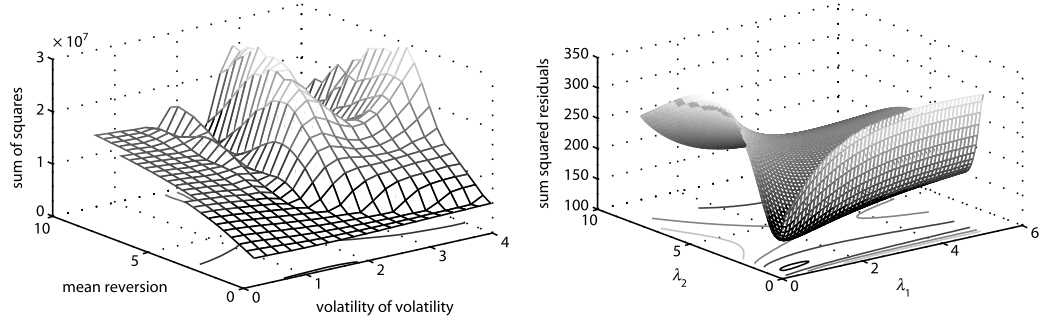


FIGURE 10.4 Left panel: Heston model objective function. Right panel: Nelson-Siegel-Svensson model objective function.

Let $y_t(\tau)$ be the yield of a zero-coupon bond with maturity τ at time t , then the Nelson-Siegel model describes the zero rates as

$$y_t(\tau) = \beta_{1,t} + \beta_{2,t} \left[\frac{1 - \exp(-\gamma_t)}{\gamma_t} \right] + \beta_{3,t} \left[\frac{1 - \exp(-\gamma_t)}{\gamma_t} - \exp(-\gamma_t) \right] \quad (10.2)$$

where $\gamma_t = \tau/\lambda_t$. The Svensson version is given by

$$y_t(\tau) = \beta_{1,t} + \beta_{2,t} \left[\frac{1 - \exp(-\gamma_{1,t})}{\gamma_{1,t}} \right] + \beta_{3,t} \left[\frac{1 - \exp(-\gamma_{1,t})}{\gamma_{1,t}} - \exp(-\gamma_{1,t}) \right] + \beta_{4,t} \left[\frac{1 - \exp(-\gamma_{2,t})}{\gamma_{2,t}} - \exp(-\gamma_{2,t}) \right] \quad (10.3)$$

where $\gamma_{1,t} = \tau/\lambda_{1,t}$ and $\gamma_{2,t} = \tau/\lambda_{2,t}$. The parameters of the models (β and λ) can be estimated by minimizing the difference between the model rates y_t and observed rates y_t^M where the superscript stands for “market.” An optimization problem could be stated as

$$\min_{\beta, \lambda} \sum (y_t - y_t^M)^2$$

subject to constraints. We need to estimate four parameters for model (10.2), and six for model (10.3). Again, this optimization problem is not convex; an example for a search space is given in Fig. 10.4. The calibration of the Nelson-Siegel model and Svensson’s extension is discussed in more detail in Chapter 16.

10.5 Summary

In this chapter, we have tried to motivate the application of optimization models in finance. The coming chapters will demonstrate how to solve such models. In Chapter 11, we will discuss classical methods: methods for zero-finding and gradient-based methods. In Chapter 12, we will give a brief introduction to heuristic methods. The emphasis of that chapter will be on principles, not on details. Later chapters will then discuss the process of solving particular problems with heuristics.